



Model-agnostic Global Feature Importance Methods

Feature Importance

Lecture: Explainable Artificial Intelligence, ST 2026

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Outline

Feature Importance

Global Model-agnostic Feature Importance Methods

- Permutation Feature Importance

- Partial Dependence Plots

- Accumulated Local Effects

Feature Interactions

Feature Importance from Surrogate Models

Summary

When do we need to be model-agnostic?

- ▶ Proprietary models, privacy constraints
- ▶ Integrated systems, non-portable execution environments (e.g., extensive GPU requirements)
- ▶ Changing model types (portability!)

Definition (Black-box)

no access to model internals (architecture, parameters, etc.)

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Definition (Feature Relevance/Importance/Attribution)

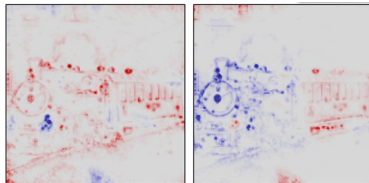
Given model $f: X_1 \times X_2 \times \dots \times X_K \rightarrow Y$, the *feature importance* of the i th input feature quantifies the influence of x_i on the output.

It answers

*What features have high **attribution** to / **influence** on*

- ▶ locally: *a specific behavior (single output)?*
- ▶ globally: *on the behavior in general (most outputs)?*

Synonyms: *feature attribution, feature relevance, feature saliency* [3].

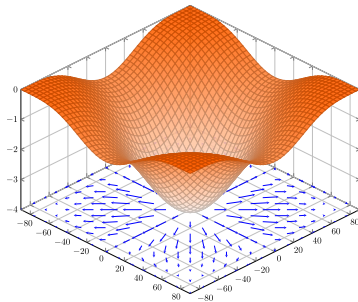


Idea

*Feature importance (vectors) point out directions of **input change** that **change model outputs strongest**.*

Problems:

- ▶ Gradients not accessible in the black-box case.
- ▶ What is a strong output change? (*absolute change, class flip, error rate, ...*)
- ▶ What is a valid input change? (*feature addition/removal/randomization, ...*)
- ▶ global: How to estimate for complete domain? (*sampling technique*)



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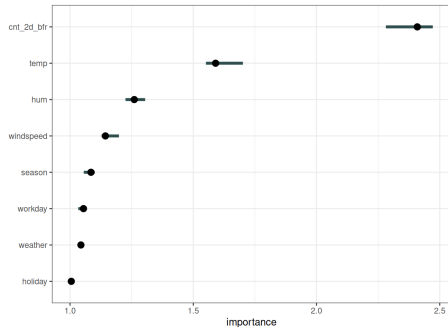
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Permutation Feature Importance

Idea: A feature is important for the problem/model when **minimizing mutual information** of it's values with the output increases the prediction error.
Minimizing: removal, setting constant, shuffling.



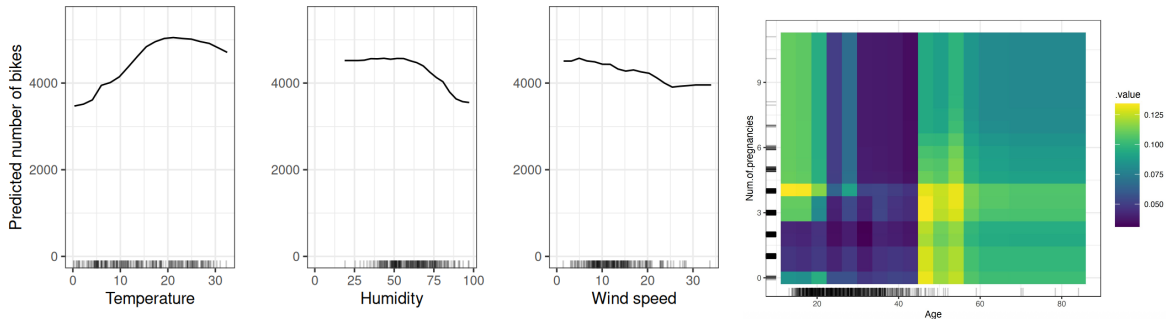
- ▶ Which features are required to solve the **problem** in general?
⇒ retrain without a feature, compare performance (used in differential testing [2]).
- ▶ Which feature is important for **this model** to maintain **performance**?
Those where shuffling does not change the average prediction error.

Pros and Cons

- + Intuitive measure for how important a feature is for the model's **performance**
- + Error ratio comparable **across different problems**
- Prone to misinterpretation: **It does not tell us anything about how changing the feature affects changing the output.**
- Random feature permutations may produce **unlikely / unrealistic** probing data points.
- Labels needed.

Global Model-agnostic Feature Importance Methods

What output to expect on average for a feature value?



Visualizing Feature Dependence with PDP

Idea: Plot how $f(x) = y$ changes on average with features $x_i, i \in S$.

Condition and then average:

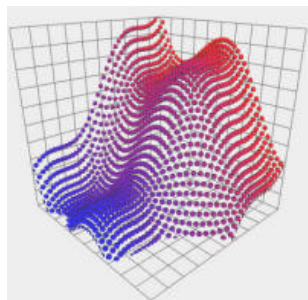
$$\text{PD}(X_S = \hat{x}_S) = \mathbb{E}_{x \sim P(X \mid X_S = \hat{x}_S)} [f(x)]$$

$$\stackrel{(*)}{\approx} \mathbb{E}_x [f(\underbrace{\hat{x}_S, x_{\neq S}}_{\text{assumed to be valid}})]$$

assumed to be valid!

$$\text{Monte Carlo} \approx \frac{1}{N} \sum_{i=1}^N f(\hat{x}_S, x_{\neq S}^{(i)})$$

(*) only holds if $X_S, X_{\neq S}$ are independent,
i.e. $P(x) = P(x_S) \cdot P(x_{\neq S})$!



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PDP Example

Assume our black-box is
 $f(x_1, x_2, x_3) = x_1 + x_3$,
and 4 samples are available:

x_1	x_2	x_3
1	1	1
1	2	2
2	2	3
2	2	4

We now *intervene* on these.

*Which feature is more
important?*

⇒ gradient idea.

$$PD(X_1 = 0) = \frac{f(0,1,1)+f(0,2,2)+f(0,2,3)+f(0,2,4)}{4} = 2.5$$

$$PD(X_1 = 1) = \frac{(f(1,1,1)+\dots)}{4} = 3.25$$

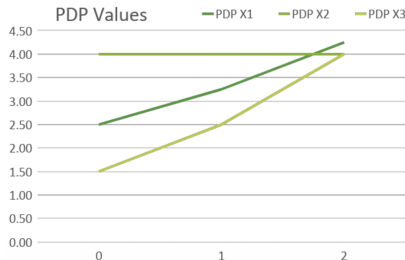
...

$$PD(X_2 = 0) = \frac{f(1,0,1)+f(1,0,2)+f(2,0,3)+f(2,0,4)}{4} = 4$$

$$PD(X_2 = 1) = \dots = 4$$

$$PD(X_2 = 2) = \dots = 4$$

...



Pros and Cons of PDP

- + Intuitive and (partly) causal: *What is the average causal effect when intervening on a feature's value?*
- Assumes **independent** features: Arbitrary interventions can lead to **unlikely samples**, e.g., *person 2m tall at 3.5kg*.
- **Averaging** over samples can cancel out effects visible for individual ones. $\Rightarrow$ individual conditional expectation (ICE) plots

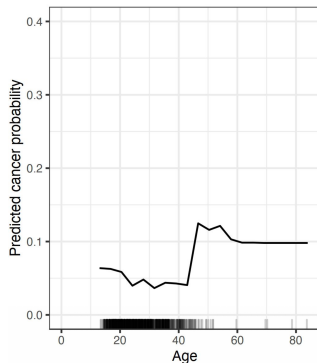
Individual Conditional Expectation (ICE) Plots

Idea: Plot the intervention curve for *each individual* sample.

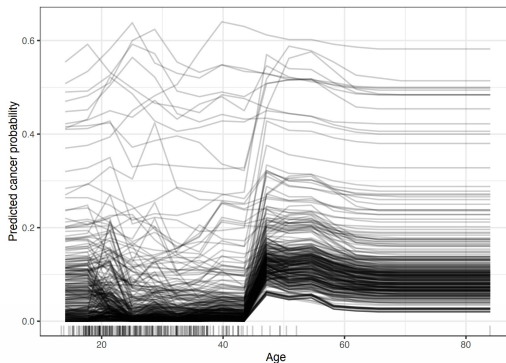
$$\text{ICE}(X_S = \hat{x}_S; \mathbf{x}) := f(\hat{x}_S, \mathbf{x}_{\neq S})$$

$$\text{PD}(X_S = \hat{x}_S) = \frac{1}{N} \sum_{i=1}^N \text{ICE}(X_S = \hat{x}_S; \mathbf{x}^{(i)})$$

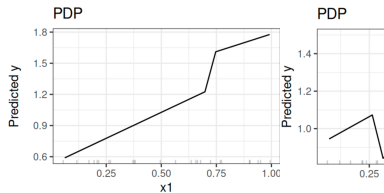
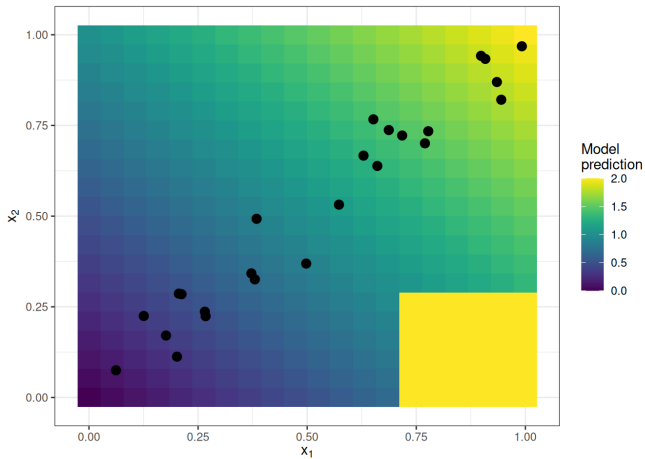
PDP



ICE



The Issue with Independence



If one strictly sticks to the data distribution for sampling (*M-Plot*):
How would the partial dependence plots (PDP) of x_1 look like for $f(x_1, x_2) = x_2$?

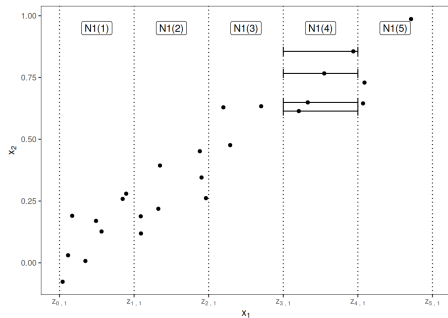
Towards Global Feature Importance of Correlated Features

Idea:

- ▶ *locally* conduct independent intervention; from that
- ▶ estimate the expected gradient $\frac{\partial f}{\partial x_k}$ (*effect*), and integrate (*accumulate*):

$$= \underbrace{\int_0^{\hat{x}_k}}_{\text{accumulate}} \underbrace{\mathbb{E}_{x \sim P(X | X_k = z_k)} \left[\frac{\partial f}{\partial x_k}(x) \right]}_{\text{local effect}} dz_k + \text{const}$$

since $g(x) = \text{const} + \int_0^x \frac{\partial g}{\partial x}(z) dz$.



Accumulated Local Effects as Estimation

1. Divide feature's range into r intervals.
2. Calculate **local effects** per interval (a, b) :
for $x \in \text{Nbh}$

2.1 Intervene to obtain **lower/upper bound** points $(a, x_{\neq k}), (b, x_{\neq k})$;

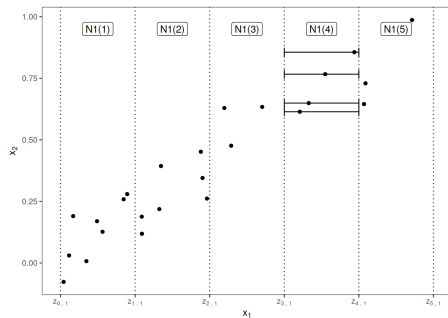
2.2 Estimate the gradient (**local effect**):

$$\begin{aligned}\text{LE}((a, b); X_k = \hat{x}_k) &:= \text{mean}_{x \in \text{Nbh}} \frac{\partial f}{\partial x_k} \left(\frac{b+a}{2}, x \right) \\ &\approx \text{mean}_{x \in \text{Nbh}} \frac{f(b, x_{\neq k}) - f(a, x_{\neq k})}{b - a}\end{aligned}$$

3. **Accumulate** to approximate integral at $x \in (a_j, b_j)$:

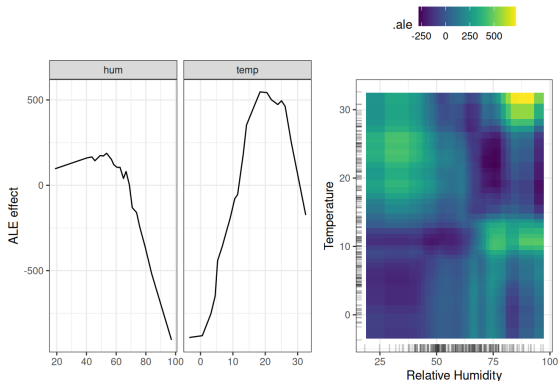
$$\text{ALE}(x; X = \hat{x}_k) := \sum_{i \leq j} \text{LE}((a_i, b_i); X = \hat{x}_k) \cdot (b_i - a_i)$$

(trick: mean-shift to center around 0)



Pros and Cons

- + Can handle correlated features better
- + More efficient than PDP
- + Still intuitive (approximates PDP)
- Strongly dependent on intervals (e.g., size)
- Unclear variance (only per interval)
- Unclear interpretation in case of feature interactions (e.g., correlation)); *changing just one of two correlated features makes little sense.*



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$x =$ This movie is not very bad. $\mapsto$ positive

↓

This movie is not very good. $\mapsto$ negative

In general: $\text{containsNot}(x) \wedge \text{containsBad}(x) \mapsto$ positive

$\Rightarrow$ We need to capture feature **interactions**!

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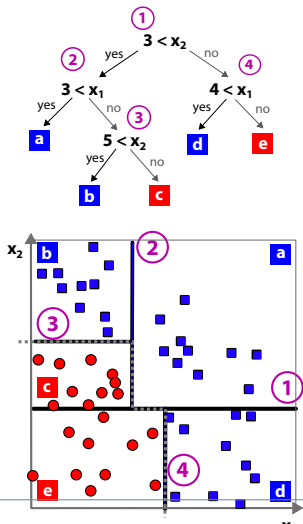
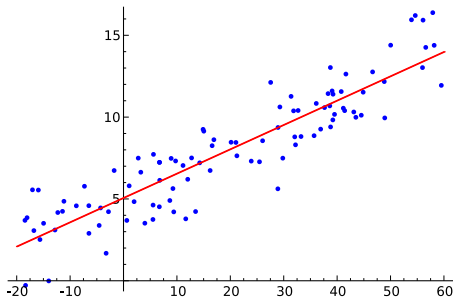
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Feature Importance from Surrogate Models

Simply approximate f via a transparent surrogate model g using $(x, f(x))$ as data for training g .



H-Statistic

Remark

No interaction:

$$\text{PD}(X_j = \hat{x}_j, X_k = \hat{x}_k) = \text{PD}(X_j = \hat{x}_j) + \text{PD}(X_k = \hat{x}_k)$$

Definition (Friedman's H-Statistic, 2-way)

For PD mean-centered at zero:

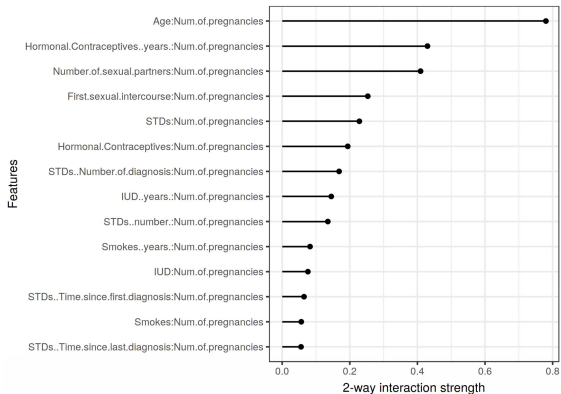
$$H_{j,k}^2 := \frac{\sum_{i=1}^N \left(\text{PD} \left(X_j = x_j^{(i)}, X_k = x_k^{(i)} \right) - \text{PD} \left(X_j = x_j^{(i)} \right) - \text{PD} \left(X_k = x_k^{(i)} \right) \right)^2}{\sum_{i=1}^N \text{PD} \left(X_j = x_j^{(i)}, X_k = x_k^{(i)} \right)^2}$$

0: no interaction,

1: all partial dependence of feature k explained by interaction.

Pros and Cons

- + Captures all kinds of feature interactions
- PDP-based: issues with unlikely combinations
- expensive



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- ▶ Permutation Feature Importance:
*Which features impact **overall performance** when shuffled?*
- ▶ PDP:
*What does the model **predict on average** if feature k has value x_k ?*
visual plot, variance available via ICE plot
- ▶ accumulated local effects (ALE): handles correlated features
- ▶ Feature interactions (H-statistic):
*How do features **mutually influence** their importance?*
- ▶ **Not ready for endusers!**

Further Reading

- ▶ Christoph Molnar, *Interpretable Machine Learning* ^[1], Chaps. 19-21, 23, 25

References

- [1] Christoph Molnar. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd ed. 2025.
- [2] Kexin Pei et al. “DeepXplore: Automated Whitebox Testing of Deep Learning Systems”. In: *Proc. 26th Symp. Operating Systems Principles*. Vol. abs/1705.06640. Shanghai, China: ACM, 2017, pp. 1–18. ISBN: 978-1-4503-5085-3. DOI: 10.1145/3132747.3132785 . arXiv: 1705.06640 .
- [3] Gesina Schwalbe and Bettina Finzel. “A Comprehensive Taxonomy for Explainable Artificial Intelligence: A Systematic Survey of Surveys on Methods and Concepts”. In: *Data Mining and Knowledge Discovery* 38 (Jan. 2023), pp. 3043–3101. ISSN: 1573-756X. DOI: 10.1007/s10618-022-00867-8 . arXiv: 2105.07190 . (Visited on 01/09/2023).